Final Phys 7120/30

(P1) $\varepsilon(\omega) = \varepsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right)$. Transverse wave with $\omega > \omega_p$.

$$\vec{\nabla}\times\vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \Rightarrow \vec{k}\times\vec{E} - \omega\vec{B} = 0.$$

$$\vec{\nabla}\times\vec{H} = \frac{\partial \vec{D}}{\partial t} \Rightarrow \vec{k}\times\vec{H} = -\omega\vec{D}$$

$$\vec{D} = \varepsilon\vec{E}, \quad \vec{B} = \mu_0\vec{H} \Rightarrow \vec{k}\times\vec{B} = -\omega\mu_0\varepsilon\vec{E}$$

$$\Rightarrow \vec{k}\times(\vec{k}\times\vec{B}) = -\omega\mu_0\varepsilon\,\vec{k}\times\vec{E} = -\omega^2\mu_0\varepsilon\,\vec{B}.$$

and $\underbrace{\vec{k}\times(\vec{k}\times\vec{E})}_{\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}} = \omega\,\vec{k}\times\vec{B} = -\omega^2\mu_0\varepsilon\,\vec{E}$

$\vec{k}\cdot\vec{E} = 0$ for transverse wave

$\Rightarrow$ wave equation is $\left(k^2 - \omega^2\mu_0\varepsilon(\omega)\right)\vec{E}(k,\omega) = 0.$

Hence $k^2 = \mu_0\varepsilon_0\frac{\varepsilon(\omega)}{\varepsilon_0}\omega^2 = \frac{1}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)\omega^2$

(a) $\Rightarrow c^2k^2 = \omega^2 - \omega_p^2$, $\omega = \sqrt{\omega_p^2 + c^2k^2}$

(b) Phase velocity $v_{ph} = \frac{\omega}{k} = \sqrt{\frac{\omega_p^2 + c^2k^2}{k^2}} =$

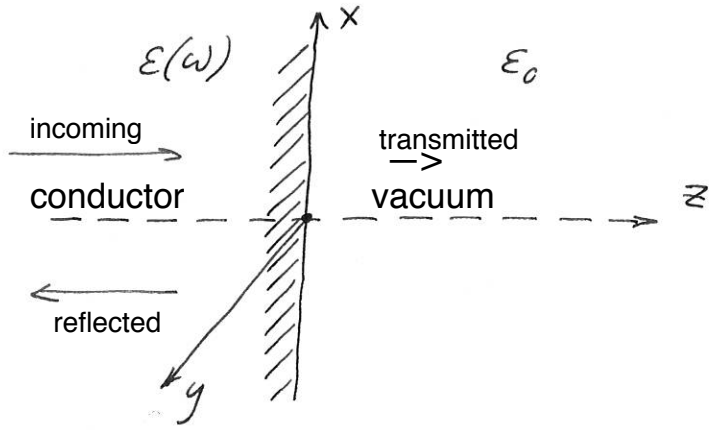
$$= c\sqrt{\frac{\omega_p^2 + c^2k^2}{c^2k^2}} = c\sqrt{\frac{\omega_p^2 + \omega^2 - \omega_p^2}{\omega^2 - \omega_p^2}} = \frac{c\,\omega}{\sqrt{\omega^2 - \omega_p^2}}$$

Note $v_{ph} \to c$ for $\omega \to \infty$; $v_g \to c$ for $\omega \to \infty$

Group velocity $v_g = \frac{\partial\omega}{\partial k} = \frac{c^2k}{\sqrt{\omega_p^2 + c^2k^2}} = c\sqrt{\frac{c^2k^2}{\omega_p^2 + c^2k^2}} =$

$= c\sqrt{\frac{\omega^2 - \omega_p^2}{\omega^2}} = c\frac{\sqrt{\omega^2 - \omega_p^2}}{\omega} < c$. Note that $v_{ph}\cdot v_g = c^2$.

Problem 2



Consider an incident wave polarized along x-direction:

$\hat{x} E_0 e^{-i\omega t + ikz}$

ω and k are

?

as $\frac{\varepsilon(\omega)}{\varepsilon_0} \equiv \epsilon_r = 1 - \frac{\omega_p^2}{\omega^2} \Rightarrow \boxed{\omega^2 - \omega_p^2 = c^2 k^2}$ (1)

Electric field (only x-components are non-zero)

is

$$E_x(z<0) = E_0 e^{-i\omega t + ikz} + E_R e^{-i\omega t - ikz} \quad (2)$$

$$E_x(z>0) = E_t e^{-i\omega t + i\tilde{k} z} \quad (3)$$

where $\tilde{k}$ is related to ω via simply $\omega = c\tilde{k}$ (4)

Magnetic field is found from

$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$ and is

along the y-direction:

$$B_y = -\frac{i}{\omega}(\nabla \times \vec{E})_y = -\frac{i}{\omega}\frac{\partial E_x}{\partial z}$$

with the help of Eqs. (2) – (3)

$$B_y(z<0) = \frac{k}{\omega} E_0 e^{-i\omega t + ikz} - \frac{k}{\omega} E_R e^{-i\omega t - ikz} \qquad (5)$$

$$B_y(z>0) = \frac{\tilde{k}}{\omega} E_t e^{-i\omega t + i\tilde{k}z} \qquad (6)$$

Continuity of E_x and B_y at $z=0$ gives

$$E_0 + E_R = E_t$$

$$k E_0 - k E_R = \tilde{k} E_t$$

which yield E_R:

$$E_R = \frac{k - \tilde{k}}{k + \tilde{k}} E_0$$

and, since both k and $\tilde{k}$ are real for $\omega > \omega_p$

$$R = \left|\frac{E_R}{E_0}\right|^2 = \left(\frac{k - \tilde{k}}{k + \tilde{k}}\right)^2$$

Using expressions for k and $\tilde{k}$ we finally obtain:

$$R = \left(\frac{\omega - \sqrt{\omega^2 - \omega_p^2}}{\omega + \sqrt{\omega^2 - \omega_p^2}}\right)^2$$

when $\omega \gg \omega_p$ the reflection coefficient approaches zero: $\sqrt{\omega^2 - \omega_p^2} \approx \omega - \frac{\omega_p^2}{2\omega}$

and $$R \approx \left(\frac{\omega_p^2/2\omega}{2\omega}\right)^2 \approx \frac{\omega_p^4}{16\omega^4}$$

When $\omega \to \omega_p$ the reflection coefficient approaches 1:

for $\omega = \omega_p - \varepsilon$ where $\varepsilon \ll \omega_p$:

$$\sqrt{\omega^2 - \omega_p^2} \approx \sqrt{2\omega_p \varepsilon}$$

and

$$R \approx \left(\frac{\omega_p - \sqrt{2\omega_p \varepsilon}}{\omega_p + \sqrt{2\omega_p \varepsilon}} \right)^2 \approx 1 - 4\sqrt{\frac{\varepsilon}{\omega_p}}$$